Operations Research 415
Part IV: Queuing Theory

Lecture 30: The Queuing System $M/G/1/GD/\infty/\infty$

Presenter: Dr Philip Venter

Department of Industrial Engineering

The System $M/G/1/GD/\infty/\infty$ (pp. 1097–1098)

Suppose the inter-arrival rate is exponentially distributed with parameter λ (and hence has expected value $\frac{1}{\lambda}$), but that the service time is *not* exponentially distributed.

Let S be the service time. Suppose that all we know about S is that $E(S) = \frac{1}{\mu}$ and var $S = \sigma^2$.

Then it can be shown that

$$\begin{aligned} L_q &= \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)} \\ L &= L_q + \rho \\ W_q &= \frac{L_q}{\lambda} \\ W &= W_q + \frac{1}{\mu} \end{aligned}$$

where $\rho = \frac{\lambda}{\mu}$.

Example (p. 1097)

Consider the system $M/M/1/GD/\infty/\infty$ with inter-arrival rate $\lambda = 5$ and service rate $\mu = 8$ as special case of the more general system $M/G/1/GD/\infty/\infty$.

1. Compute the quantities L, L_q, W and W_q when viewing the queuing system as an $M/M/1/GD/\infty/\infty$ system.
2. Now reevaluate the quantities L, L_q, W and W_q when viewing the queuing system as an $M/G/1/GD/\infty/\infty$ system.

Problem 3 (p. 1098)

An average of 40 cars per hour arrive to be painted at a single-server GM painting facility.

95% of the cars require 1 minute to paint, while 5% must be painted twice and require 2.5 minutes to paint.

Assume that interarrival times are exponential.

1. On average, how long does a car wait before being painted?
2. If cars never had to be repainted, how would your answer to part (1) change?